Alexandra Kashani Motlagh

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Summary

Software engineer with a B.S. in Computer Science & Engineering from The Ohio State University. Published first-author researcher (Springer Nature) with two NSF-funded research experiences in AI/ML, NLP, and cloud infrastructure analysis. Proficient in Python, Java, and JavaScript with hands-on experience building full-stack applications, LLM-integrated tooling, and interactive 3D environments.

Education

The Ohio State University — Columbus, OH
B.S. in Computer Science & Engineering — Computer Graphics & Game Design — *Aug 2019 – May 2025*

- **Scholarships:** Bill & Sue Lhota Endowed (2023–2025) · Wening Memorial (2024–2025)
- **Coursework:** Data Structures & Algorithms, Operating Systems, Software Engineering, Database Systems, AI & Machine Learning, OOP

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C#, C/C++, SQL, HTML5, CSS3
Frameworks & Libraries: React, Three.js, Node.js, Django, WebGL, OpenGL
Developer Tools: Git, GitHub, Docker, Linux/Unix, REST APIs, CI/CD, VS Code
Concepts: OOP, SDLC, Agile/Scrum, Microservices, Unit Testing, Cloud Computing (AWS)
AI/ML: Natural Language Processing, LLM Integration & Evaluation, Infrastructure-as-Code Analysis

Research Experience

University of Arizona & University of Oulu — Oulu, Finland
NSF IRES Undergraduate Researcher, Systems & Industrial Engineering — *May 2024 – Aug 2024*

- Selected for competitive NSF IRES Track I program: U.S.–Finnish research on sustainable cloud-native systems and technical debt management
- Analyzed **390+ open-source GitHub repositories** from the MSR Dataset to identify Infrastructure-as-Code usage patterns across Terraform, Ansible, and 11 additional IaC services
- Built automated Python filter-parsing tools for 13 IaC platforms and integrated **7 LLM models** (GPT-4, LLaMA, Mistral) for repository classification, improving categorization speed and accuracy
- Contributed to an LLM-based system for automated cloud infrastructure generation, applying prompt engineering and output evaluation techniques

University of Texas at Arlington — Arlington, TX
NSF REU Undergraduate Researcher, Computer Science & Engineering — *Jun 2023 – Aug 2023*

- Conducted research on real-time speech-to-ASL translation under Dr. Ishfaq Ahmad and Dr. Mari F. Tietze
- Proposed a pioneering real-time ASL gesture projection solution using NLP models; optimized the ASL-Gloss pipeline to reduce latency and improve translation accuracy
- First-author publication in **Springer Nature** at the World Congress in Computer Science & Applied Computing, 2024

Projects

Infrastructure-as-Code in Microservices Dataset | *Python, AI/ML, LLM APIs* — 2024

- Developed custom Python tools to classify IaC usage across 390 GitHub repos for 13 service platforms; validated results by querying 7 LLM APIs, achieving consistent cross-model agreement

IttyBityCity — 3D Procedural City Explorer | *Three.js, JavaScript, WebGL* — 2026

- Built an interactive 3D city in the browser with procedural building generation, dynamic camera controls, and real-time rendering using Three.js and WebGL shaders

Sims Game Remake — Browser Life Simulation | *JavaScript, HTML5/CSS3, OOP* — 2026

- Developed a browser-based life simulation using OOP design patterns, event-driven architecture, and DOM manipulation; deployed and playable online

RushRoute — Procedural Obstacle Game | *JavaScript, HTML5 Canvas* — 2024

- Created a fast-paced obstacle-avoidance game with procedural level generation, progressive difficulty scaling, and frame-rate-independent game loop

Publication

Kashani Motlagh, A. (First Author), Mehta, S., Ahmad, I., Clark, A., Lei, H. *A Performance Comparison Between Two Speech-To-ASL Gesture-Projection Translation Implementations.* Springer Nature, 2024.

Additional Experience & Leadership

Key Holder, Retail Operations — Dublin, OH — Managed daily operations, cash reconciliation, and team coordination
Laboratory Monitor, The Ohio State University — Columbus, OH — Technical support for engineering computer labs
Society of Women Engineers, Member — 2021 – 2025